

IK solver rotates the bones automatically. A character's legs are a perfect place to use IK because it allows the hips to move freely while the feet remain firmly planted on the ground.



Positioning characters legs is a perfect place to use IK

6.3 Working with IK

IK can be applied to a bone chain as it is drawn, or it can be applied afterward by applying an IK solver. Choose Animation > IK Solvers, and then the desired solver. Then select the start and the end bone of the IK chain. 3ds Max has three main IK solvers, plus the ability to do Spline IK. The three main solvers are as follows:

- **HI solver**

The History Independent solver uses an IK goal to animate a chain. You animate the goal, and the solver moves the end of the chain to match the position of the goal. This is the preferred solver for most applications.

- **HD solver**

The History Dependent solver lets you use sliding joints combined with inverse kinematics. It has controls for spring back, damping, and precedence not found in the HI solver. The solver, however, is history dependent; performance is slower at the end of long animations.

- **IK Limb solver**

The IK Limb solver is meant for animating the limbs of characters; for example, the hip to the ankle, or the shoulder to the wrist. Each IK Limb solver affects only two bones, but multiple solvers can be applied to different parts of the same chain. IK chains are configured by selecting the IK goal and opening the Motion panel.

6.3.1 Using Spline IK

Another way to configure IK is by using Spline IK, which employs a curve to control the bones and their rotations. This is a good choice for manipulating long chains such as a tail, a trunk, or even a spine. Spline IK works best with chains that have a lot of short bones. To create a Spline IK chain, draw a bone chain with the Spline IK solver enabled. 3ds Max automatically generates a curve of your choice to fit the chain, plus it gives you the option to create helper objects to manipulate the underlying curve and the chain. Once